Mathematical Methods

Module 3 – Exam Exercise Pack

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Probability spaces, random variables, distributions, convergence, characteristic functions and conditional expectation

How to use this exercise pack

Method: Exam training strategy

Each exercise has a difficulty level from $\star$ to $\star\star\star\star\star$ and a detailed solution. For each problem, first identify the mathematical object: event, random variable, distribution, expectation, variance, conditional expectation or convergence statement.

- Try the problem before reading the solution.
- Rewrite every probability as the probability of an event.
- For random variables, first find the support and then compute probabilities.
- For convergence questions, state clearly which convergence is being used.

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Conditional probability, Bayes formula and independence

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Convergence and limit theorems

Characteristic functions

Conditional expectation

Module 3 exam training

Probability spaces and events

Exercise 001 – Sample space from a finite experiment

Exercise: Difficulty: $\star$

A box contains the symbols a, b, c, d. One symbol is selected uniformly. Define $X(\omega)$ as 1 if the selected symbol is a vowel and 0 otherwise. Write Ω, $\mathcal{F}$, X, $\mathbb{P}(X = 1)$ and $\mathbb{E}[X]$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 001 – Sample space from a finite experiment

Solution: Detailed solution

Step 1: the possible outcomes are $\Omega = \{a, b, c, d\}$. Since the space is finite, take $\mathcal{F} = 2^{\Omega}$. Step 2: only a is a vowel, so $X(a) = 1$ and $X(b) = X(c) = X(d) = 0$. Step 3: $\{X = 1\} = \{a\}$, hence $\mathbb{P}(X = 1) = 1/4$. Step 4: since X is an indicator, $\mathbb{E}[X] = \mathbb{P}(X = 1) = 1/4$.

Exercise 002 – Events generated by a random variable

Exercise: Difficulty: $\star\star$

Roll two fair dice and let S be their sum. Write the event $\{S \leq 4\}$ explicitly as a subset of $\Omega = \{1, \ldots, 6\}^2$ and compute its probability.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 002 – Events generated by a random variable

Solution: Detailed solution

The event is made of all ordered pairs with sum at most 4:

$$\{S \leq 4\} = \{(1,1), (1,2), (2,1), (1,3), (2,2), (3,1)\}.$$

There are 36 equally likely outcomes. Therefore

$$\mathbb{P}(S \leq 4) = \frac{6}{36} = \frac{1}{6}.$$

Exercise 003 – Sigma-algebra check

Exercise: Difficulty: ⋆⋆

Let $\Omega = \{1, 2, 3, 4\}$ and $\mathcal{F} = \{\varnothing, \Omega, \{1, 2\}, \{3, 4\}\}$. Is $\mathcal{F}$ a sigma-algebra? Justify every property.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 003 – Sigma-algebra check

Solution: Detailed solution

First, $\varnothing, \Omega \in \mathcal{F}$. Complements are in $\mathcal{F}$: $\{1,2\}^c = \{3,4\}$ and $\{3,4\}^c = \{1,2\}$. Unions of elements of $\mathcal{F}$ give only $\varnothing$, $\{1,2\}$, $\{3,4\}$ or Ω, all in $\mathcal{F}$. Since the family is finite, closure under finite unions implies closure under countable unions. Hence $\mathcal{F}$ is a sigma-algebra.

Exercise 004 – Smallest sigma-algebra

Exercise: Difficulty: $\star\star$

Let $\Omega = \{1, 2, 3, 4, 5, 6\}$ and $A = \{1, 3, 5\}$. Find the smallest sigma-algebra containing A.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 004 – Smallest sigma-algebra

Solution: Detailed solution

A sigma-algebra containing A must contain $A^c = \{2, 4, 6\}$, also $\varnothing$ and Ω. These four sets already form a sigma-algebra. Therefore

$$\sigma(A) = \{\varnothing, \Omega, A, A^c\}.$$

Exercise 005 – Probability measure on a non-uniform finite space

Exercise: Difficulty: ⋆⋆⋆

Let $\Omega = \{\omega_1, \omega_2, \omega_3\}$ and define $\mathbb{P}(\{\omega_1\}) = p$, $\mathbb{P}(\{\omega_2\}) = 2p$, $\mathbb{P}(\{\omega_3\}) = 1 - 3p$. For which values of p is this a probability measure?

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 005 – Probability measure on a non-uniform finite space

Solution: Detailed solution

A probability on a finite space is determined by nonnegative masses summing to one. The sum is

$$p + 2p + (1 - 3p) = 1,$$

so total mass is always one. Nonnegativity requires $p \geq 0$, $2p \geq 0$, and $1 - 3p \geq 0$. Thus

$$0 \leq p \leq \frac{1}{3}.$$

Exercise 006 – Inclusion-exclusion with numbers

Exercise: Difficulty: $\star\star$

Suppose $\mathbb{P}(A) = 0.62$, $\mathbb{P}(B) = 0.48$ and $\mathbb{P}(A \cap B) = 0.25$. Compute $\mathbb{P}(A \cup B)$, $\mathbb{P}(A \setminus B)$ and $\mathbb{P}(B \setminus A)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 006 – Inclusion-exclusion with numbers

Solution: Detailed solution

Use inclusion-exclusion:

$$\mathbb{P}(A \cup B) = 0.62 + 0.48 - 0.25 = 0.85.$$

Next remove the overlap:

$$\mathbb{P}(A \setminus B) = \mathbb{P}(A) - \mathbb{P}(A \cap B) = 0.37,$$
$$\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A \cap B) = 0.23.$$

Exercise 007 – Bounds for an intersection

Exercise: Difficulty: ⋆⋆⋆

Assume $\mathbb{P}(A) = 0.35$ and $\mathbb{P}(B) = 0.80$. Find the smallest and largest possible value of $\mathbb{P}(A \cap B)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 007 – Bounds for an intersection

Solution: Detailed solution

The intersection cannot exceed the smaller event, so

$$\mathbb{P}(A \cap B) \leq \min\{0.35, 0.80\} = 0.35.$$

Also $\mathbb{P}(A \cup B) \leq 1$. Since $\mathbb{P}(A \cup B) = 0.35 + 0.80 - \mathbb{P}(A \cap B)$,

$$1.15 - \mathbb{P}(A \cap B) \leq 1,$$

so $\mathbb{P}(A \cap B) \geq 0.15$. Therefore

$$0.15 \leq \mathbb{P}(A \cap B) \leq 0.35.$$

Exercise 008 – Continuity from below

Exercise: Difficulty: $\star\star$

Let $A_n = \{1, 2, \ldots, n\}$ inside $\Omega = \mathbb{N}$ and suppose $\mathbb{P}(\{k\}) = 2^{-k}$. Compute $\lim_{n\to\infty} \mathbb{P}(A_n)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 008 – Continuity from below

Solution: Detailed solution

The events increase: $A_n \uparrow \mathbb{N}$. By continuity from below,

$$\lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}(\mathbb{N}).$$

Now $\sum_{k=1}^{\infty} 2^{-k} = 1$, so $\mathbb{P}(\mathbb{N}) = 1$. Hence the limit is 1.

Exercise 009 – Continuity from above

Exercise: Difficulty: ⋆⋆⋆

Let X be a finite-valued random variable and define $A_n = \{|X| > n\}$. Show that $\mathbb{P}(A_n) \to 0$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 009 – Continuity from above

Solution: Detailed solution

The events decrease because $\{|X| > n+1\} \subseteq \{|X| > n\}$. Since X is finite-valued, for every outcome ω there exists n such that $|X(\omega)| \leq n$. Hence

$$\bigcap_{n=1}^{\infty} A_n = \varnothing.$$

By continuity from above,

$$\mathbb{P}(A_n) \to \mathbb{P}(\varnothing) = 0.$$

Exercise 010 – Proving a basic consequence

Exercise: Difficulty: ⋆⋆⋆

Starting from the axioms of probability, prove that $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 010 – Proving a basic consequence

Solution: Detailed solution

The sets A and A^c are disjoint and their union is Ω. By additivity,

$$\mathbb{P}(\Omega) = \mathbb{P}(A \cup A^c) = \mathbb{P}(A) + \mathbb{P}(A^c).$$

Since $\mathbb{P}(\Omega) = 1$, rearranging gives

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

Exercise 011 – A counterexample about additivity

Exercise: Difficulty: ⋆⋆⋆⋆

Give an explicit probability space and two events A, B for which $\mathbb{P}(A \cup B) \neq \mathbb{P}(A) + \mathbb{P}(B)$. Explain exactly why the false formula fails.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 011 – A counterexample about additivity

Solution: Detailed solution

Take a fair die. Let $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$. Then

$$\mathbb{P}(A) = \frac{1}{2}, \qquad \mathbb{P}(B) = \frac{1}{2},$$

but $A \cup B = \{1, 2, 3, 4, 5\}$, so

$$\mathbb{P}(A \cup B) = \frac{5}{6} \neq 1.$$

The false formula counts the overlap $A \cap B = \{3\}$ twice. The correct formula subtracts $\mathbb{P}(A \cap B)$.

Exercise 012 – Events and measurability

Exercise: Difficulty: ⋆⋆⋆⋆

Let $\Omega = \{1, 2, 3, 4\}$ and $\mathcal{F} = \{\varnothing, \Omega, \{1, 2\}, \{3, 4\}\}$. Define $X(1) = X(3) = 0$ and $X(2) = X(4) = 1$. Is X measurable with respect to $\mathcal{F}$?

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 012 – Events and measurability

Solution: Detailed solution

For a finite real-valued variable, check inverse images of singletons. We have

$$X^{-1}(\{0\}) = \{1, 3\}, \qquad X^{-1}(\{1\}) = \{2, 4\}.$$

Neither set belongs to $\mathcal{F}$. Therefore X is not measurable. The point is that a formula can be written down but still fail to be an observable random variable relative to the chosen sigma-algebra.

Module 3 exam training

Conditional probability, Bayes formula and independence

Exercise 013 – Conditional probability from a table

Exercise: Difficulty: ⋆⋆

In a class of 40 students, 18 study topic A, 25 study topic B, and 12 study both. A student is selected uniformly. Compute $\mathbb{P}(A \mid B)$ and $\mathbb{P}(B \mid A)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 013 – Conditional probability from a table

Solution: Detailed solution

Use $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$. Since probabilities are counts divided by 40,

$$\mathbb{P}(A \mid B) = \frac{12/40}{25/40} = \frac{12}{25}.$$

Similarly,

$$\mathbb{P}(B \mid A) = \frac{12/40}{18/40} = \frac{12}{18} = \frac{2}{3}.$$

Exercise 014 – Bayes formula with two causes

Exercise: Difficulty: ⋆⋆

A machine is selected from line L_1 with probability 0.7 and from line L_2 with probability 0.3. The probability of a defect is 0.02 on L_1 and 0.05 on L_2. Given that the item is defective, compute the probability it came from L_2.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 014 – Bayes formula with two causes

Solution: Detailed solution

Let D be the defect event. First compute total probability:

$$\mathbb{P}(D) = 0.02 \cdot 0.7 + 0.05 \cdot 0.3 = 0.014 + 0.015 = 0.029.$$

Bayes formula gives

$$\mathbb{P}(L_2 \mid D) = \frac{\mathbb{P}(D \mid L_2)\mathbb{P}(L_2)}{\mathbb{P}(D)} = \frac{0.05 \cdot 0.3}{0.029} = \frac{0.015}{0.029} \approx 0.5172.$$

Exercise 015 – Independence check

Exercise: Difficulty: ⋆⋆⋆

Let $\mathbb{P}(A) = 0.4$, $\mathbb{P}(B) = 0.5$, and $\mathbb{P}(A \cup B) = 0.7$. Are A and B independent?

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 015 – Independence check

Solution: Detailed solution

First find the intersection:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B) = 0.4 + 0.5 - 0.7 = 0.2.$$

If A and B were independent, we would have

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) = 0.4 \cdot 0.5 = 0.2.$$

The equality holds, so A and B are independent.

Exercise 016 – Independent complements

Exercise: Difficulty: ⋆⋆⋆

Prove that if A and B are independent, then A^c and B are independent.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 016 – Independent complements

Solution: Detailed solution

We need to prove $\mathbb{P}(A^c \cap B) = \mathbb{P}(A^c)\mathbb{P}(B)$. Since $B = (A \cap B) \cup (A^c \cap B)$ disjointly,

$$\mathbb{P}(A^c \cap B) = \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

By independence, $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. Hence

$$\mathbb{P}(A^c \cap B) = \mathbb{P}(B)[1 - \mathbb{P}(A)] = \mathbb{P}(B)\mathbb{P}(A^c).$$

Exercise 017 – Pairwise versus mutual independence

Exercise: Difficulty: ⋆⋆⋆⋆

Toss two fair coins. Let $A =$ "first toss is heads", $B =$ "second toss is heads", and $C =$ "the two tosses are equal". Show that A, B, C are pairwise independent but not mutually independent.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 017 – Pairwise versus mutual independence

Solution: Detailed solution

Each event has probability $1/2$. Also $A \cap B = \{HH\}$, $A \cap C = \{HH\}$, and $B \cap C = \{HH\}$, each with probability $1/4$, equal to the product $1/2 \cdot 1/2$. Thus they are pairwise independent. But

$$A \cap B \cap C = \{HH\}, \qquad \mathbb{P}(A \cap B \cap C) = \frac{1}{4},$$

whereas

$$\mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C) = \frac{1}{8}.$$

Therefore they are not mutually independent.

Exercise 018 – Conditional probability as a new probability

Exercise: Difficulty: ⋆⋆⋆

Let B be an event with $\mathbb{P}(B) > 0$. Define $Q(A) = \mathbb{P}(A \mid B)$. Prove that Q is a probability measure.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 018 – Conditional probability as a new probability

Solution: Detailed solution

First $Q(A) \geq 0$ because probabilities are nonnegative. Next

$$Q(\Omega) = \mathbb{P}(\Omega \mid B) = \frac{\mathbb{P}(B)}{\mathbb{P}(B)} = 1.$$

If (A_n) are disjoint, then $(A_n \cap B)$ are disjoint. Therefore

$$Q\left(\bigcup_n A_n\right) = \frac{\mathbb{P}\left((\cup_n A_n) \cap B\right)}{\mathbb{P}(B)} = \frac{\sum_n \mathbb{P}(A_n \cap B)}{\mathbb{P}(B)} = \sum_n Q(A_n).$$

So Q satisfies the axioms.

Exercise 019 – At least one success by conditioning

Exercise: Difficulty: ⋆⋆

An event has probability $p = 0.15$ in each of 8 independent trials. Compute the probability that it occurs at least once.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 019 – At least one success by conditioning

Solution: Detailed solution

The complement of "at least once" is "never". By independence,

$$\mathbb{P}(\text{never}) = (1-p)^8 = 0.85^8.$$

Therefore

$$\mathbb{P}(\text{at least once}) = 1 - 0.85^8 \approx 1 - 0.2725 = 0.7275.$$

Exercise 020 – Bayes with three causes

Exercise: Difficulty: ⋆⋆⋆

A signal comes from source S_1, S_2, S_3 with probabilities $0.5, 0.35, 0.15$. The probabilities of alarm are $0.01, 0.04, 0.10$. Given an alarm, compute the posterior probability of S_3.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 020 – Bayes with three causes

Solution: Detailed solution

Let A be the alarm. Total probability gives

$$\mathbb{P}(A) = 0.01(0.5) + 0.04(0.35) + 0.10(0.15) = 0.005 + 0.014 + 0.015 = 0.034.$$

Then

$$\mathbb{P}(S_3 \mid A) = \frac{\mathbb{P}(A \mid S_3)\mathbb{P}(S_3)}{\mathbb{P}(A)} = \frac{0.10 \cdot 0.15}{0.034} = \frac{0.015}{0.034} \approx 0.4412.$$

Exercise 021 – Conditional independence trap

Exercise: Difficulty: ⋆⋆⋆

Let two fair coins be tossed. Let $A =$ "first is heads", $B =$ "second is heads", and $C =$ "at least one head". Are A and B independent conditionally on C?

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 021 – Conditional independence trap

Solution: Detailed solution

Given C, the possible outcomes are HH, HT, TH, each with conditional probability $1/3$. Thus

$$\mathbb{P}(A \mid C) = \frac{2}{3}, \qquad \mathbb{P}(B \mid C) = \frac{2}{3}.$$

Also

$$\mathbb{P}(A \cap B \mid C) = \mathbb{P}(HH \mid C) = \frac{1}{3}.$$

But $(2/3)(2/3) = 4/9 \neq 1/3$. Hence A and B are not conditionally independent given C.

Exercise 022 – Law of total probability from a partition

Exercise: Difficulty: ⋆⋆⋆⋆

Let B_1, B_2, B_3 be a partition with positive probabilities. Prove that for every event A,

$$\mathbb{P}(A) = \sum_{i=1}^{3} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 022 – Law of total probability from a partition

Solution: Detailed solution

Since the B_i form a partition,

$$A = (A \cap B_1) \cup (A \cap B_2) \cup (A \cap B_3),$$

and the union is disjoint. By additivity,

$$\mathbb{P}(A) = \sum_{i=1}^{3} \mathbb{P}(A \cap B_i).$$

Using $\mathbb{P}(A \cap B_i) = \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)$ gives the desired formula.

Exercise 023 – Independence from conditional probability

Exercise: Difficulty: ⋆⋆

Suppose $\mathbb{P}(A) = 0.3$, $\mathbb{P}(B) = 0.6$ and $\mathbb{P}(A \mid B) = 0.3$. Are A and B independent?

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 023 – Independence from conditional probability

Solution: Detailed solution

If $\mathbb{P}(B) > 0$, independence is equivalent to $\mathbb{P}(A \mid B) = \mathbb{P}(A)$. Here

$$\mathbb{P}(A \mid B) = 0.3 = \mathbb{P}(A).$$

Therefore A and B are independent. Equivalently, $\mathbb{P}(A \cap B) = 0.3 \cdot 0.6 = 0.18$.

Exercise 024 – Finding a missing conditional probability

Exercise: Difficulty: ⋆⋆⋆⋆

Let $\mathbb{P}(A) = 0.45$, $\mathbb{P}(B) = 0.50$, and $\mathbb{P}(A \mid B) = 0.30$. Compute $\mathbb{P}(B \mid A)$ and decide whether A, B are independent.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 024 – Finding a missing conditional probability

Solution: Detailed solution

First compute the intersection:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B) = 0.30 \cdot 0.50 = 0.15.$$

Then

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} = \frac{0.15}{0.45} = \frac{1}{3}.$$

Independence would require $\mathbb{P}(A \cap B) = 0.45 \cdot 0.50 = 0.225$, which is false. Hence they are not independent.

Exercise 025 – Conditioning with cards

Exercise: Difficulty: ⋆⋆⋆

A card is drawn uniformly from a standard 52-card deck. Compute the probability that it is an ace given that it is red, and the probability that it is red given that it is an ace.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 025 – Conditioning with cards

Solution: Detailed solution

There are 26 red cards and 2 red aces. Thus

$$\mathbb{P}(\text{ace} \mid \text{red}) = \frac{2}{26} = \frac{1}{13}.$$

There are 4 aces and 2 are red, so

$$\mathbb{P}(\text{red} \mid \text{ace}) = \frac{2}{4} = \frac{1}{2}.$$

The two conditional probabilities answer different questions and need not coincide.

Exercise 026 – A counterexample to transitivity of independence

Exercise: Difficulty: ⋆⋆⋆⋆⋆

Construct events A, B, C such that A is independent of B and B is independent of C, but A is not independent of C.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 026 – A counterexample to transitivity of independence

Solution: Detailed solution

Use one fair coin and let $A = C = \{H\}$, while $B = \Omega$. Then B is independent of every event because

$$\mathbb{P}(A \cap \Omega) = \mathbb{P}(A) = \mathbb{P}(A)\mathbb{P}(\Omega).$$

Thus A is independent of B and B is independent of C. However A and C are the same nontrivial event, so

$$\mathbb{P}(A \cap C) = \mathbb{P}(A) = 1/2 \neq 1/4 = \mathbb{P}(A)\mathbb{P}(C).$$

Thus independence is not transitive.

Module 3 exam training

Discrete random variables and probability models

Exercise 027 – Probability mass function

Exercise: Difficulty: $\star$

Let X have mass function $\mathbb{P}(X = 0) = 0.2$, $\mathbb{P}(X = 1) = 0.5$, $\mathbb{P}(X = 3) = 0.3$. Compute $\mathbb{E}[X]$ and $\mathbb{P}(X \geq 1)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 027 – Probability mass function

Solution: Detailed solution

Use the definition of expectation for a discrete variable:

$$\mathbb{E}[X] = 0(0.2) + 1(0.5) + 3(0.3) = 0.5 + 0.9 = 1.4.$$

The event $\{X \geq 1\}$ is $\{X = 1\} \cup \{X = 3\}$, hence

$$\mathbb{P}(X \geq 1) = 0.5 + 0.3 = 0.8.$$

Exercise 028 – Finding a normalizing constant

Exercise: Difficulty: ⋆⋆

Let $\mathbb{P}(X = k) = c(k + 1)$ for $k = 0, 1, 2, 3$. Find c and compute $\mathbb{E}[X]$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 028 – Finding a normalizing constant

Solution: Detailed solution

Normalize:

$$1 = \sum_{k=0}^{3} c(k+1) = c(1+2+3+4) = 10c,$$

so $c = 1/10$. Then

$$\mathbb{E}[X] = \sum_{k=0}^{3} k\frac{k+1}{10} = 0 + \frac{2}{10} + \frac{6}{10} + \frac{12}{10} = 2.$$

Exercise 029 – Bernoulli variable

Exercise: Difficulty: $\star\star$

Let $X \sim \text{Bern}(p)$. Derive $\mathbb{E}[X]$ and $\text{Var}(X)$ from the definition.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 029 – Bernoulli variable

Solution: Detailed solution

Since X takes values 0 and 1,

$$\mathbb{E}[X] = 0(1-p) + 1p = p.$$

Also $X^2 = X$, so $\mathbb{E}[X^2] = p$. Therefore

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1-p).$$

Exercise 030 – Binomial probability

Exercise: Difficulty: ⋆⋆

Let $X \sim \text{Bin}(5, 0.3)$. Compute $\mathbb{P}(X = 2)$ and $\mathbb{P}(X \leq 1)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 030 – Binomial probability

Solution: Detailed solution

The binomial mass is

$$\mathbb{P}(X = k) = \binom{5}{k}(0.3)^k(0.7)^{5-k}.$$

Thus

$$\mathbb{P}(X = 2) = \binom{5}{2}(0.3)^2(0.7)^3 = 10(0.09)(0.343) = 0.3087.$$

Also

$$\mathbb{P}(X \leq 1) = 0.7^5 + 5(0.3)(0.7)^4 \approx 0.16807 + 0.36015 = 0.52822.$$

Exercise 031 – Poisson recursion

Exercise: Difficulty: ⋆⋆⋆

Let $X \sim \text{Poisson}(4)$. Use the recursion $\mathbb{P}(X = k+1) = \frac{4}{k+1}\mathbb{P}(X = k)$ to compute $\mathbb{P}(X = 3)$ from $\mathbb{P}(X = 0)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 031 – Poisson recursion

Solution: Detailed solution

Start from $\mathbb{P}(X=0)=e^{-4}$. Then

$$\mathbb{P}(X=1)=4e^{-4},$$

$$\mathbb{P}(X=2)=\frac{4}{2}\mathbb{P}(X=1)=8e^{-4},$$

$$\mathbb{P}(X=3)=\frac{4}{3}\mathbb{P}(X=2)=\frac{32}{3}e^{-4}.$$

This agrees with $e^{-4}4^3/3!$.

Exercise 032 – Distribution function of a discrete variable

Exercise: Difficulty: ⋆⋆⋆

Let X take values $-1, 0, 2$ with probabilities $0.25, 0.50, 0.25$. Write the distribution function $F_X(x)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 032 – Distribution function of a discrete variable

Solution: Detailed solution

The distribution function is $F_X(x) = \mathbb{P}(X \leq x)$. Thus

$$F_X(x) = \begin{cases} 0, & x < -1, \\ 0.25, & -1 \leq x < 0, \\ 0.75, & 0 \leq x < 2, \\ 1, & x \geq 2. \end{cases}$$

The jumps occur exactly at the possible values of X.

Exercise 033 – Expectation of a transformed variable

Exercise: Difficulty: ⋆⋆⋆

Let X have $\mathbb{P}(X=-2)=0.2$, $\mathbb{P}(X=1)=0.5$, $\mathbb{P}(X=4)=0.3$. Compute $\mathbb{E}[(X-1)^2]$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 033 – Expectation of a transformed variable

Solution: Detailed solution

Use the law of the unconscious statistician:

$$\mathbb{E}[(X-1)^2] = \sum_x (x-1)^2 \mathbb{P}(X=x).$$

Therefore

$$(-3)^2(0.2) + 0^2(0.5) + 3^2(0.3) = 9(0.2) + 0 + 9(0.3) = 4.5.$$

Exercise 034 – Geometric waiting time

Exercise: Difficulty: ⋆⋆⋆

Let X be the number of trials until the first success, with success probability $p = 0.2$. Compute $\mathbb{P}(X > 4)$ and interpret the result.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 034 – Geometric waiting time

Solution: Detailed solution

The event $\{X > 4\}$ means no success in the first four trials. By independence,

$$\mathbb{P}(X > 4) = (1 - p)^4 = 0.8^4 = 0.4096.$$

So there is about a 40.96% probability that one waits more than four trials.

Exercise 035 – Memoryless property

Exercise: Difficulty: ⋆⋆⋆⋆

Let X be geometric with $\mathbb{P}(X = k) = (1-p)^{k-1}p$, $k \geq 1$. Prove that $\mathbb{P}(X > m+n \mid X > m) = \mathbb{P}(X > n)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 035 – Memoryless property

Solution: Detailed solution

For the geometric waiting time,

$$\mathbb{P}(X > r) = (1-p)^r.$$

Therefore

$$\mathbb{P}(X > m+n \mid X > m) = \frac{\mathbb{P}(X > m+n)}{\mathbb{P}(X > m)} = \frac{(1-p)^{m+n}}{(1-p)^m} = (1-p)^n.$$

But $(1-p)^n = \mathbb{P}(X > n)$, proving the memoryless property.

Exercise 036 – Indicator decomposition

Exercise: Difficulty: ⋆⋆⋆

Let X be the number of heads in 10 fair coin tosses. Write X as a sum of indicators and compute $\mathbb{E}[X]$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 036 – Indicator decomposition

Solution: Detailed solution

Let $I_i = 1$ if toss i is heads, and 0 otherwise. Then

$$X = I_1 + \cdots + I_{10}.$$

Each $I_i \sim \text{Bern}(1/2)$, so $\mathbb{E}[I_i] = 1/2$. By linearity of expectation,

$$\mathbb{E}[X] = \sum_{i=1}^{10} \mathbb{E}[I_i] = 10 \cdot \frac{1}{2} = 5.$$

No independence is needed for linearity, although it is true here.

Exercise 037 – Variance of a binomial variable

Exercise: Difficulty: ⋆⋆⋆⋆

Using indicators, derive the variance of $X \sim \mathrm{Bin}(n, p)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 037 – Variance of a binomial variable

Solution: Detailed solution

Write $X = \sum_{i=1}^{n} I_i$, where the I_i are independent Bernoulli(p) variables. Then

$$\mathrm{Var}(I_i) = p(1-p).$$

Because the indicators are independent, covariances vanish, so

$$\mathrm{Var}(X) = \sum_{i=1}^{n} \mathrm{Var}(I_i) = np(1-p).$$

Exercise 038 – A non-binomial sum

Exercise: Difficulty: ⋆⋆⋆⋆

Let I_1, I_2 be Bernoulli variables with $\mathbb{P}(I_1 = 1) = \mathbb{P}(I_2 = 1) = 1/2$ and $I_2 = I_1$. Is $X = I_1 + I_2$ binomial $\mathrm{Bin}(2, 1/2)$?

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 038 – A non-binomial sum

Solution: Detailed solution

No. If $I_2 = I_1$, then X takes only values 0 and 2:

$$\mathbb{P}(X = 0) = 1/2, \qquad \mathbb{P}(X = 2) = 1/2,$$

and $\mathbb{P}(X = 1) = 0$. A $\text{Bin}(2, 1/2)$ variable would have $\mathbb{P}(X = 1) = 1/2$. The issue is that binomial sums require independent Bernoulli trials.

Exercise 039 – Poisson approximation

Exercise: Difficulty: ⋆⋆⋆

A rare event occurs in each of 200 independent trials with probability 0.01. Approximate the probability of exactly 3 occurrences using a Poisson approximation.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 039 – Poisson approximation

Solution: Detailed solution

Here $X \sim \text{Bin}(200, 0.01)$ and $np = 2$. For small p and large n, approximate $X \approx \text{Poisson}(2)$. Then

$$\mathbb{P}(X = 3) \approx e^{-2}\frac{2^3}{3!} = e^{-2}\frac{8}{6} = \frac{4}{3}e^{-2} \approx 0.1804.$$

Exercise 040 – Distribution of a maximum

Exercise: Difficulty: ⋆⋆⋆

Let X_1, X_2, X_3 be independent with $\mathbb{P}(X_i \leq x) = F(x)$. Find the distribution function of $M = \max(X_1, X_2, X_3)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 040 – Distribution of a maximum

Solution: Detailed solution

Use the event equivalence

$$\{M \le x\} = \{X_1 \le x, X_2 \le x, X_3 \le x\}.$$

By independence,

$$\mathbb{P}(M \le x) = \prod_{i=1}^{3} \mathbb{P}(X_i \le x) = F(x)^3.$$

Thus $F_M(x) = F(x)^3$.

Exercise 041 – Distribution of a minimum

Exercise: Difficulty: ⋆⋆⋆⋆

Let $X_1, \ldots, X_n$ be independent with common distribution function F. Find $\mathbb{P}(\min_i X_i > x)$ and the distribution function of the minimum.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 041 – Distribution of a minimum

Solution: Detailed solution

Let $m = \min_i X_i$. Then

$$\{m > x\} = \{X_1 > x, \ldots, X_n > x\}.$$

By independence,

$$\mathbb{P}(m > x) = [1 - F(x)]^n.$$

Therefore

$$F_m(x) = \mathbb{P}(m \leq x) = 1 - [1 - F(x)]^n.$$

Exercise 042 – A probability generating function

Exercise: Difficulty: ⋆⋆⋆⋆⋆

Let X have probability generating function $G_X(s) = 0.2 + 0.5s + 0.3s^3$. Find the support, $\mathbb{E}[X]$ and $\mathbb{P}(X = 2)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 042 – A probability generating function

Solution: Detailed solution

The coefficient of s^k is $\mathbb{P}(X = k)$. Hence the support is $\{0, 1, 3\}$ with probabilities $0.2, 0.5, 0.3$. There is no s^2 term, so $\mathbb{P}(X = 2) = 0$. Also

$$G_X'(s) = 0.5 + 0.9s^2,$$

so

$$\mathbb{E}[X] = G_X'(1) = 0.5 + 0.9 = 1.4.$$

Module 3 exam training

Continuous random variables and transformations

Exercise 043 – Density normalization

Exercise: Difficulty: $\star\star$

Let $f(x) = cx$ on $0 < x < 2$ and $f(x) = 0$ otherwise. Find c and compute $\mathbb{P}(X > 1)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 043 – Density normalization

Solution: Detailed solution

Normalize the density:

$$1 = \int_0^2 cx\,dx = c\frac{x^2}{2}\Big|_0^2 = 2c,$$

so $c = 1/2$. Then

$$\mathbb{P}(X > 1) = \int_1^2 \frac{x}{2}\,dx = \frac{x^2}{4}\Big|_1^2 = 1 - \frac{1}{4} = \frac{3}{4}.$$

Exercise 044 – Distribution function from a density

Exercise: Difficulty: ⋆⋆

Let X have density $f(x) = 2x$ on $0 < x < 1$. Compute $F_X(x)$ for all real x.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 044 – Distribution function from a density

Solution: Detailed solution

For $x \le 0$, $F_X(x) = 0$. For $0 < x < 1$,

$$F_X(x) = \int_0^x 2t\,dt = x^2.$$

For $x \ge 1$, $F_X(x) = 1$. Thus

$$F_X(x) = \begin{cases} 0, & x \le 0, \\ x^2, & 0 < x < 1, \\ 1, & x \ge 1. \end{cases}$$

Exercise 045 – Expectation from a density

Exercise: Difficulty: $\star\star\star$

Let X have density $f(x) = 3x^2$ on $0 < x < 1$. Compute $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 045 – Expectation from a density

Solution: Detailed solution

Compute moments:

$$\mathbb{E}[X] = \int_0^1 x\, 3x^2\, dx = 3\int_0^1 x^3 dx = \frac{3}{4}.$$

Also

$$\mathbb{E}[X^2] = \int_0^1 x^2\, 3x^2\, dx = 3\int_0^1 x^4 dx = \frac{3}{5}.$$

Therefore

$$\mathrm{Var}(X) = \frac{3}{5} - \left(\frac{3}{4}\right)^2 = \frac{48-45}{80} = \frac{3}{80}.$$

Exercise 046 – Uniform transformation

Exercise: Difficulty: ⋆⋆⋆

Let $U \sim \text{Unif}(0,1)$ and define $Y = -\log U$. Find the distribution of Y.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 046 – Uniform transformation

Solution: Detailed solution

For $y < 0$, $F_Y(y) = 0$. For $y \geq 0$,

$$F_Y(y) = \mathbb{P}(-\log U \leq y) = \mathbb{P}(\log U \geq -y) = \mathbb{P}(U \geq e^{-y}).$$

Since U is uniform,

$$\mathbb{P}(U \geq e^{-y}) = 1 - e^{-y}.$$

Thus Y has density e^{-y} on $y \geq 0$, so $Y \sim \text{Exp}(1)$.

Exercise 047 – Exponential survival

Exercise: Difficulty: ⋆⋆⋆

Let $X \sim \mathrm{Exp}(\lambda)$ with density $\lambda e^{-\lambda x}$ for $x \geq 0$. Compute $\mathbb{P}(X > s + t \mid X > s)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 047 – Exponential survival

Solution: Detailed solution

The survival function is

$$\mathbb{P}(X > x) = \int_x^\infty \lambda e^{-\lambda u}\, du = e^{-\lambda x}.$$

Therefore

$$\mathbb{P}(X > s + t \mid X > s) = \frac{\mathbb{P}(X > s + t)}{\mathbb{P}(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t}.$$

This equals $\mathbb{P}(X > t)$.

Exercise 048 – Change of variables

Exercise: Difficulty: ⋆⋆⋆⋆

Let X have density $f_X(x) = 2x$ on $(0, 1)$ and define $Y = X^2$. Find the density of Y.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 048 – Change of variables

Solution: Detailed solution

For $0 < y < 1$, the transformation is $y = x^2$, so $x = \sqrt{y}$ and

$$\frac{dx}{dy} = \frac{1}{2\sqrt{y}}.$$

Thus

$$f_Y(y) = f_X(\sqrt{y}) \left| \frac{dx}{dy} \right| = 2\sqrt{y} \cdot \frac{1}{2\sqrt{y}} = 1.$$

Hence $Y \sim \text{Unif}(0, 1)$.

Exercise 049 – Density of an absolute value

Exercise: Difficulty: ⋆⋆⋆⋆

Let X have density $f_X(x) = \frac{1}{2}$ on $(-1, 1)$. Find the density of $Y = |X|$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 049 – Density of an absolute value

Solution: Detailed solution

For $0 < y < 1$, the value $Y = y$ can arise from $X = y$ or $X = -y$. Therefore

$$f_Y(y) = f_X(y) + f_X(-y) = \frac{1}{2} + \frac{1}{2} = 1.$$

So Y has density 1 on $(0, 1)$, i.e. $Y \sim \text{Unif}(0, 1)$.

Exercise 050 – Normal standardization

Exercise: Difficulty: $\star\star\star$

Let $X \sim \mathcal{N}(10, 4)$, where 4 is the variance. Express $\mathbb{P}(8 < X < 13)$ in terms of the standard normal distribution function Φ.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 050 – Normal standardization

Solution: Detailed solution

The standard deviation is 2. Standardize:

$$Z = \frac{X - 10}{2} \sim \mathcal{N}(0, 1).$$

The inequalities become

$$8 < X < 13 \quad \Longleftrightarrow \quad -1 < Z < 1.5.$$

Hence

$$\mathbb{P}(8 < X < 13) = \Phi(1.5) - \Phi(-1).$$

Exercise 051 – Quantile calculation

Exercise: Difficulty: ⋆⋆⋆⋆

Let $X \sim \text{Exp}(2)$. Find the median m, i.e. the number satisfying $\mathbb{P}(X \leq m) = 1/2$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 051 – Quantile calculation

Solution: Detailed solution

The distribution function is $F(x) = 1 - e^{-2x}$ for $x \geq 0$. Set $F(m) = 1/2$:

$$1 - e^{-2m} = \frac{1}{2}.$$

Thus $e^{-2m} = 1/2$. Taking logarithms,

$$-2m = -\log 2,$$

so

$$m = \frac{\log 2}{2}.$$

Exercise 052 – Mixed distribution warning

Exercise: Difficulty: ⋆⋆⋆⋆

Let X satisfy $\mathbb{P}(X = 0) = 1/3$ and, conditional on $X > 0$, have density $2e^{-2x}$ on $x > 0$. Write $F_X(x)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 052 – Mixed distribution warning

Solution: Detailed solution

For $x < 0$, $F_X(x) = 0$. At $x = 0$, there is an atom:

$$F_X(0) = \mathbb{P}(X = 0) = \frac{1}{3}.$$

For $x > 0$,

$$F_X(x) = \frac{1}{3} + \mathbb{P}(0 < X \leq x).$$

The continuous part has total mass $2/3$, so

$$\mathbb{P}(0 < X \leq x) = \frac{2}{3}(1 - e^{-2x}).$$

Thus

$$F_X(x) = \frac{1}{3} + \frac{2}{3}(1 - e^{-2x}), \quad x > 0.$$

Exercise 053 – Tail integral formula

Exercise: Difficulty: ⋆⋆⋆⋆

Let $X \geq 0$ be continuous with survival function $\mathbb{P}(X > x) = e^{-x^2}$ for $x \geq 0$. Express $\mathbb{E}[X]$ using the tail integral formula.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 053 – Tail integral formula

Solution: Detailed solution

For a nonnegative random variable,

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > x)\, dx.$$

Therefore

$$\mathbb{E}[X] = \int_0^\infty e^{-x^2}\, dx = \frac{\sqrt{\pi}}{2}.$$

The last equality is the standard Gaussian integral over the positive half-line.

Exercise 054 – Finding a density from a CDF

Exercise: Difficulty: ⋆⋆⋆⋆⋆

Let

$$F(x) = \begin{cases} 0, & x < 0, \\ x^3, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

Show that F is a distribution function and find the density.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 054 – Finding a density from a CDF

Solution: Detailed solution

The function is nondecreasing, right-continuous, with limits 0 at $-\infty$ and 1 at $+\infty$, so it is a distribution function. On $(0,1)$ it is differentiable and

$$f(x) = F'(x) = 3x^2.$$

There are no jumps at 0 or 1 because $F(0) = 0$ and $F(1^-) = 1 = F(1)$. Hence the density is $3x^2\mathbf{1}_{(0,1)}(x)$.

Exercise 055 – Nonlinear transformation with two roots

Exercise: Difficulty: ⋆⋆⋆⋆⋆

Let $X \sim \text{Unif}(-1, 1)$ and $Y = X^2$. Find $F_Y(y)$ and $f_Y(y)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 055 – Nonlinear transformation with two roots

Solution: Detailed solution

For $y < 0$, $F_Y(y) = 0$. For $0 \le y \le 1$,

$$F_Y(y) = \mathbb{P}(X^2 \le y) = \mathbb{P}(-\sqrt{y} \le X \le \sqrt{y}).$$

Since X is uniform on an interval of length 2,

$$F_Y(y) = \frac{2\sqrt{y}}{2} = \sqrt{y}.$$

For $y \ge 1$, $F_Y(y) = 1$. Differentiating on $(0, 1)$ gives

$$f_Y(y) = \frac{1}{2\sqrt{y}}.$$

Module 3 exam training

Expectation, variance and inequalities

Exercise 056 – Linearity of expectation

Exercise: Difficulty: ⋆⋆

Let $\mathbb{E}[X] = 2$ and $\mathbb{E}[Y] = -1$. Compute $\mathbb{E}[3X - 4Y + 5]$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 056 – Linearity of expectation

Solution: Detailed solution

Linearity of expectation gives

$$\mathbb{E}[3X - 4Y + 5] = 3\mathbb{E}[X] - 4\mathbb{E}[Y] + 5.$$

Substitute the given values:

$$3(2) - 4(-1) + 5 = 6 + 4 + 5 = 15.$$

No independence assumption is needed.

Exercise 057 – Variance scaling

Exercise: Difficulty: ⋆⋆

Let $\mathrm{Var}(X) = 9$. Compute $\mathrm{Var}(2 - 3X)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 057 – Variance scaling

Solution: Detailed solution

Adding a constant does not change variance, and multiplying by a constant a multiplies variance by a^2. Thus

$$\mathrm{Var}(2 - 3X) = (-3)^2 \,\mathrm{Var}(X) = 9 \cdot 9 = 81.$$

Exercise 058 – Variance of a sum with covariance

Exercise: Difficulty: ⋆⋆⋆

Suppose $\mathrm{Var}(X) = 4$, $\mathrm{Var}(Y) = 9$, and $\mathrm{Cov}(X, Y) = -2$. Compute $\mathrm{Var}(X + Y)$ and $\mathrm{Var}(2X - Y)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 058 – Variance of a sum with covariance

Solution: Detailed solution

Use

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y) = 4 + 9 - 4 = 9.$$

For $2X - Y$,

$$\mathrm{Var}(2X - Y) = 4\,\mathrm{Var}(X) + \mathrm{Var}(Y) - 4\,\mathrm{Cov}(X,Y).$$

Therefore

$$4(4) + 9 - 4(-2) = 16 + 9 + 8 = 33.$$

Exercise 059 – Jensen inequality

Exercise: Difficulty: ⋆⋆⋆

Let $X > 0$ and $\mathbb{E}[X] = 4$. Use Jensen's inequality to bound $\mathbb{E}[\log X]$ from above.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 059 – Jensen inequality

Solution: Detailed solution

The function $\log x$ is concave. Jensen's inequality for concave functions gives

$$\mathbb{E}[\log X] \leq \log(\mathbb{E}[X]).$$

Since $\mathbb{E}[X] = 4$,

$$\mathbb{E}[\log X] \leq \log 4.$$

Equality holds only in the degenerate case where $X = 4$ almost surely.

Exercise 060 – Markov inequality

Exercise: Difficulty: ⋆⋆⋆

Let $X \geq 0$ and $\mathbb{E}[X] = 3$. Give an upper bound for $\mathbb{P}(X \geq 10)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 060 – Markov inequality

Solution: Detailed solution

Markov's inequality states

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}, \qquad a > 0.$$

With $a = 10$,

$$\mathbb{P}(X \geq 10) \leq \frac{3}{10} = 0.3.$$

Exercise 061 – Chebyshev inequality

Exercise: Difficulty: ⋆⋆⋆

Let $\mathbb{E}[X] = 5$ and $\mathrm{Var}(X) = 16$. Bound $\mathbb{P}(|X - 5| \geq 8)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 061 – Chebyshev inequality

Solution: Detailed solution

Chebyshev's inequality gives

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}.$$

Here $a = 8$, so

$$\mathbb{P}(|X - 5| \geq 8) \leq \frac{16}{64} = \frac{1}{4}.$$

Exercise 062 – Cauchy-Schwarz for covariance

Exercise: Difficulty: ⋆⋆⋆⋆

Prove that $|\operatorname{Cov}(X, Y)| \leq \sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}$ when the variances are finite.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 062 – Cauchy-Schwarz for covariance

Solution: Detailed solution

Set $U = X - \mathbb{E}[X]$ and $V = Y - \mathbb{E}[Y]$. Then $\mathrm{Cov}(X, Y) = \mathbb{E}[UV]$. By Cauchy-Schwarz,

$$|\mathbb{E}[UV]| \le \sqrt{\mathbb{E}[U^2]}\sqrt{\mathbb{E}[V^2]}.$$

Since $\mathbb{E}[U^2] = \mathrm{Var}(X)$ and $\mathbb{E}[V^2] = \mathrm{Var}(Y)$,

$$|\,\mathrm{Cov}(X, Y)| \le \sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}.$$

Exercise 063 – A variance identity

Exercise: Difficulty: ⋆⋆⋆⋆

Show that $\mathbb{E}[(X-a)^2]$ is minimized over $a \in \mathbb{R}$ at $a = \mathbb{E}[X]$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 063 – A variance identity

Solution: Detailed solution

Write

$$X - a = (X - \mathbb{E}[X]) + (\mathbb{E}[X] - a).$$

Squaring and taking expectation gives

$$\mathbb{E}[(X-a)^2] = \mathbb{E}[(X-\mathbb{E}[X])^2] + 2(\mathbb{E}[X]-a)\mathbb{E}[X-\mathbb{E}[X]] + (\mathbb{E}[X]-a)^2.$$

The middle term is zero. Hence

$$\mathbb{E}[(X-a)^2] = \mathrm{Var}(X) + (a - \mathbb{E}[X])^2,$$

which is minimized when $a = \mathbb{E}[X]$.

Exercise 064 – Expectation of an indicator

Exercise: Difficulty: ⋆ ⋆ ⋆

Let A be an event. Prove that $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$ and use this to compute $\mathbb{E}[3\mathbf{1}_A - 2\mathbf{1}_{A^c}]$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 064 – Expectation of an indicator

Solution: Detailed solution

The indicator $\mathbf{1}_A$ equals 1 on A and 0 otherwise, so

$$\mathbb{E}[\mathbf{1}_A] = 1 \cdot \mathbb{P}(A) + 0 \cdot \mathbb{P}(A^c) = \mathbb{P}(A).$$

By linearity,

$$\mathbb{E}[3\mathbf{1}_A - 2\mathbf{1}_{A^c}] = 3\mathbb{P}(A) - 2\mathbb{P}(A^c).$$

Since $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$, this equals

$$5\mathbb{P}(A) - 2.$$

Exercise 065 – Computing moments from tails

Exercise: Difficulty: ⋆⋆⋆⋆

Let $X \geq 0$ and suppose $\mathbb{P}(X > x) = 1/(1+x)^2$ for $x \geq 0$. Compute $\mathbb{E}[X]$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 065 – Computing moments from tails

Solution: Detailed solution

Use the tail formula:

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > x)\,dx = \int_0^\infty \frac{1}{(1+x)^2}\,dx.$$

Since

$$\int \frac{1}{(1+x)^2}\,dx = -\frac{1}{1+x},$$

we get

$$\mathbb{E}[X] = \left[-\frac{1}{1+x}\right]_0^\infty = 0 - (-1) = 1.$$

Exercise 066 – A variable with no finite expectation

Exercise: Difficulty: ⋆⋆⋆⋆

Let X have density $f(x) = 1/x^2$ on $x \geq 1$. Show that f is a density but $\mathbb{E}[X] = \infty$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 066 – A variable with no finite expectation

Solution: Detailed solution

First normalize:

$$\int_1^\infty \frac{1}{x^2}\,dx = \left[-\frac{1}{x}\right]_1^\infty = 1,$$

so f is a density. But

$$\mathbb{E}[X] = \int_1^\infty x\frac{1}{x^2}\,dx = \int_1^\infty \frac{1}{x}\,dx = \infty.$$

Thus a valid random variable can fail to have finite expectation.

Exercise 067 – Uniform integrability warning

Exercise: Difficulty: ⋆⋆⋆⋆⋆

Let $X_n = n\mathbf{1}_{(0,1/n)}$ on $((0,1), \mathcal{B}(0,1), \lambda)$. Show that $X_n \to 0$ in probability but $\mathbb{E}[X_n] = 1$ for all n.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 067 – Uniform integrability warning

Solution: Detailed solution

For any $\varepsilon > 0$, if $n > \varepsilon$, then

$$\mathbb{P}(|X_n| > \varepsilon) = \lambda(0, 1/n) = \frac{1}{n} \to 0.$$

Hence $X_n \to 0$ in probability. However

$$\mathbb{E}[X_n] = \int_0^{1/n} n \, dx = 1.$$

Therefore convergence in probability alone does not imply convergence of expectations.

Module 3 exam training

Random vectors, joint distributions and covariance

Exercise 068 – Joint mass table

Exercise: Difficulty: ⋆⋆

Let (X, Y) have joint probabilities $p(0,0) = 0.2$, $p(0,1) = 0.3$, $p(1,0) = 0.1$, $p(1,1) = 0.4$. Find the marginal distributions of X and Y.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 068 – Joint mass table

Solution: Detailed solution

For X, sum over Y:

$$\mathbb{P}(X=0) = 0.2 + 0.3 = 0.5, \qquad \mathbb{P}(X=1) = 0.1 + 0.4 = 0.5.$$

For Y, sum over X:

$$\mathbb{P}(Y=0) = 0.2 + 0.1 = 0.3, \qquad \mathbb{P}(Y=1) = 0.3 + 0.4 = 0.7.$$

Exercise 069 – Checking independence from a joint table

Exercise: Difficulty: ⋆⋆⋆

Using the joint distribution in the previous exercise, decide whether X and Y are independent.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 069 – Checking independence from a joint table

Solution: Detailed solution

If X, Y were independent, then for every pair (x, y),

$$p(x, y) = \mathbb{P}(X = x)\mathbb{P}(Y = y).$$

From the marginals, $\mathbb{P}(X = 0) = 0.5$ and $\mathbb{P}(Y = 0) = 0.3$, so independence would imply

$$p(0, 0) = 0.5 \cdot 0.3 = 0.15.$$

But the table gives $p(0, 0) = 0.2$. Hence X and Y are not independent.

Exercise 070 – Covariance from a joint table

Exercise: Difficulty: ⋆⋆⋆

Let (X, Y) take values $(0,0), (0,2), (1,0), (1,2)$ with probabilities $0.25, 0.25, 0.10, 0.40$. Compute $\mathrm{Cov}(X, Y)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 070 – Covariance from a joint table

Solution: Detailed solution

First compute expectations:

$$\mathbb{E}[X] = 0(0.25 + 0.25) + 1(0.10 + 0.40) = 0.50.$$

$$\mathbb{E}[Y] = 0(0.25 + 0.10) + 2(0.25 + 0.40) = 1.30.$$

Next

$$\mathbb{E}[XY] = 0 + 0 + 0 + 1 \cdot 2 \cdot 0.40 = 0.80.$$

Thus

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0.80 - 0.50(1.30) = 0.15.$$

Exercise 071 – Correlation calculation

Exercise: Difficulty: ⋆⋆⋆

Suppose $\mathrm{Var}(X) = 9$, $\mathrm{Var}(Y) = 16$, and $\mathrm{Cov}(X, Y) = 6$. Compute $\mathrm{Corr}(X, Y)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 071 – Correlation calculation

Solution: Detailed solution

Correlation is covariance divided by the product of standard deviations:

$$\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)}\sqrt{\mathrm{Var}(Y)}}.$$

Therefore

$$\mathrm{Corr}(X, Y) = \frac{6}{3 \cdot 4} = \frac{1}{2}.$$

Exercise 072 – Uncorrelated but dependent

Exercise: Difficulty: ⋆⋆⋆⋆

Let X take values $-1, 0, 1$ with probabilities $1/3$ each, and define $Y = X^2$. Show that X and Y are uncorrelated but dependent.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 072 – Uncorrelated but dependent

Solution: Detailed solution

We have $\mathbb{E}[X] = 0$ by symmetry. Also $XY = X^3$, and

$$\mathbb{E}[X^3] = \frac{-1 + 0 + 1}{3} = 0.$$

Thus $\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0$. They are uncorrelated. But they are dependent because Y is determined by X: if $X = 0$ then $Y = 0$, while if $X = \pm 1$ then $Y = 1$.

Exercise 073 – Joint density normalization

Exercise: Difficulty: ⋆⋆⋆

Let $f(x,y) = c(x+y)$ on $0 < x < 1$, $0 < y < 1$, and 0 otherwise. Find c.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 073 – Joint density normalization

Solution: Detailed solution

Normalize over the unit square:

$$1 = c \int_0^1 \int_0^1 (x + y)\, dy\, dx.$$

Compute the inner integral:

$$\int_0^1 (x + y) dy = x + \frac{1}{2}.$$

Then

$$\int_0^1 \left(x + \frac{1}{2}\right) dx = \frac{1}{2} + \frac{1}{2} = 1.$$

Hence $c = 1$.

Exercise 074 – Marginal density

Exercise: Difficulty: ⋆⋆⋆

For the joint density $f(x, y) = x + y$ on $(0, 1)^2$, find the marginal density of X.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 074 – Marginal density

Solution: Detailed solution

Integrate out y:

$$f_X(x) = \int_0^1 (x + y)\, dy = x + \frac{1}{2}, \qquad 0 < x < 1.$$

Outside $(0, 1)$ the marginal density is zero. Check normalization:

$$\int_0^1 \left(x + \frac{1}{2}\right) dx = 1.$$

Exercise 075 – Conditional density

Exercise: Difficulty: ⋆⋆⋆⋆⋆

Let $f(x,y) = x + y$ on $(0,1)^2$. Find the conditional density of Y given $X = x$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 075 – Conditional density

Solution: Detailed solution

The conditional density is

$$f_{Y|X=x}(y) = \frac{f(x,y)}{f_X(x)}.$$

From the previous calculation, $f_X(x) = x + 1/2$. Therefore, for $0 < y < 1$,

$$f_{Y|X=x}(y) = \frac{x+y}{x+1/2}.$$

It integrates to one because the denominator is precisely $\int_0^1 (x+y)dy$.

Exercise 076 – Expectation of a sum

Exercise: Difficulty: $\star\star\star$

Let X, Y be any integrable random variables. Prove $\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y]$ in the discrete finite case.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 076 – Expectation of a sum

Solution: Detailed solution

Let (X, Y) take finitely many values (x_i, y_i) with probabilities p_i. Then

$$\mathbb{E}[X + Y] = \sum_i (x_i + y_i) p_i.$$

Splitting the sum gives

$$\sum_i x_i p_i + \sum_i y_i p_i = \mathbb{E}[X] + \mathbb{E}[Y].$$

This proof shows why no independence assumption is involved.

Exercise 077 – Variance of a linear combination

Exercise: Difficulty: ⋆⋆⋆⋆

Let X, Y have $\mathrm{Var}(X) = 1$, $\mathrm{Var}(Y) = 4$, and $\mathrm{Cov}(X, Y) = 1$. Compute $\mathrm{Var}(3X + 2Y)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 077 – Variance of a linear combination

Solution: Detailed solution

Use

$$\operatorname{Var}(aX + bY) = a^2 \operatorname{Var}(X) + b^2 \operatorname{Var}(Y) + 2ab \operatorname{Cov}(X, Y).$$

With $a = 3$, $b = 2$,

$$\operatorname{Var}(3X + 2Y) = 9(1) + 4(4) + 2(3)(2)(1) = 9 + 16 + 12 = 37.$$

Exercise 078 – Independence via density factorization

Exercise: Difficulty: ⋆⋆⋆⋆⋆

Let $f(x,y) = 6xy^2$ on $0 < x < 1$, $0 < y < 1$. Show that X and Y are independent and find their marginal densities.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 078 – Independence via density factorization

Solution: Detailed solution

Compute marginals:

$$f_X(x) = \int_0^1 6xy^2\,dy = 6x \cdot \frac{1}{3} = 2x,$$

for $0 < x < 1$, and

$$f_Y(y) = \int_0^1 6xy^2\,dx = 6y^2 \cdot \frac{1}{2} = 3y^2,$$

for $0 < y < 1$. Their product is

$$f_X(x)f_Y(y) = (2x)(3y^2) = 6xy^2 = f(x, y).$$

Hence X and Y are independent.

Exercise 079 – Transforming independent variables

Exercise: Difficulty: ⋆⋆⋆⋆⋆

Let X, Y be independent standard normal variables. Define $U = X + Y$ and $V = X - Y$. Compute $\mathrm{Cov}(U, V)$ and state what follows if (U, V) is jointly normal.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 079 – Transforming independent variables

Solution: Detailed solution

Compute

$$\mathrm{Cov}(U, V) = \mathrm{Cov}(X + Y, X - Y).$$

Expanding gives

$$\mathrm{Var}(X) - \mathrm{Cov}(X, Y) + \mathrm{Cov}(Y, X) - \mathrm{Var}(Y).$$

Since X, Y are independent standard normals, $\mathrm{Cov}(X, Y) = 0$ and both variances are 1. Hence

$$\mathrm{Cov}(U, V) = 1 - 1 = 0.$$

Because linear transformations of a jointly normal vector are jointly normal, U, V are jointly normal. For jointly normal variables, zero covariance implies independence.

Exercise 080 – Conditional expectation from a joint table

Exercise: Difficulty: ⋆⋆⋆⋆⋆

Let (X, Y) have probabilities $p(0,0) = 0.1$, $p(0,1) = 0.4$, $p(2,0) = 0.3$, $p(2,1) = 0.2$. Compute $\mathbb{E}[X \mid Y = 1]$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 080 – Conditional expectation from a joint table

Solution: Detailed solution

First compute $\mathbb{P}(Y = 1) = 0.4 + 0.2 = 0.6$. Conditional probabilities are

$$\mathbb{P}(X = 0 \mid Y = 1) = \frac{0.4}{0.6} = \frac{2}{3},$$

$$\mathbb{P}(X = 2 \mid Y = 1) = \frac{0.2}{0.6} = \frac{1}{3}.$$

Therefore

$$\mathbb{E}[X \mid Y = 1] = 0 \cdot \frac{2}{3} + 2 \cdot \frac{1}{3} = \frac{2}{3}.$$

Module 3 exam training

Convergence and limit theorems

Exercise 081 – Sum of independent Poisson variables

Exercise: Difficulty: $\star\star\star\star\star$

Let $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ be independent. Use probability generating functions to find the distribution of $X + Y$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 081 – Sum of independent Poisson variables

Solution: Detailed solution

The probability generating function of a Poisson variable is

$$G_X(s) = \exp(\lambda(s-1)), \qquad G_Y(s) = \exp(\mu(s-1)).$$

Independence implies

$$G_{X+Y}(s) = G_X(s)G_Y(s) = \exp((\lambda+\mu)(s-1)).$$

This is the generating function of a Poisson$(\lambda+\mu)$ variable. Thus

$$X + Y \sim \text{Poisson}(\lambda+\mu).$$

Exercise 082 – Convergence in probability

Exercise: Difficulty: ⋆ ⋆ ⋆

Let $X_n = \mathbf{1}_{(0,1/n)}$ on $((0,1), \mathcal{B}(0,1), \lambda)$. Show that $X_n \to 0$ in probability.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 082 – Convergence in probability

Solution: Detailed solution

For any $\varepsilon \in (0,1)$,

$$\mathbb{P}(|X_n - 0| > \varepsilon) = \mathbb{P}(X_n = 1) = \lambda(0, 1/n) = \frac{1}{n}.$$

Since $1/n \to 0$, we have $X_n \to 0$ in probability.

Exercise 083 – Almost sure convergence

Exercise: Difficulty: ⋆⋆⋆

Let $X_n(\omega) = \omega^n$ on $\Omega = (0,1)$ with Lebesgue probability. Show that $X_n \to 0$ almost surely.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 083 – Almost sure convergence

Solution: Detailed solution

For every fixed $\omega \in (0,1)$, the deterministic sequence ω^n tends to 0. Hence the set of outcomes where convergence fails is empty. Therefore

$$X_n \to 0 \quad \text{almost surely.}$$

Exercise 084 – Probability but not almost sure

Exercise: Difficulty: ⋆⋆⋆⋆

Let $X_n = \mathbf{1}_{A_n}$ on $[0,1]$, where A_n cycles through dyadic intervals in the usual typewriter sequence. Explain why $X_n \to 0$ in probability but not almost surely.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 084 – Probability but not almost sure

Solution: Detailed solution

At each stage the intervals become shorter, so $\lambda(A_n) \to 0$. Hence, for $0 < \varepsilon < 1$,

$$\mathbb{P}(|X_n| > \varepsilon) = \lambda(A_n) \to 0,$$

so $X_n \to 0$ in probability. However every point $\omega \in [0, 1]$ belongs to infinitely many of the cycling intervals. Thus $X_n(\omega) = 1$ infinitely often and 0 infinitely often, so the pointwise limit does not exist. Hence there is no almost sure convergence.

Exercise 085 – L^2 implies probability

Exercise: Difficulty: ⋆⋆⋆

Suppose $\mathbb{E}[(X_n - X)^2] \to 0$. Prove that $X_n \to X$ in probability.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 085 – L^2 implies probability

Solution: Detailed solution

Use Markov's inequality applied to the nonnegative variable $(X_n - X)^2$:

$$\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{P}((X_n - X)^2 > \varepsilon^2) \leq \frac{\mathbb{E}[(X_n - X)^2]}{\varepsilon^2}.$$

The numerator tends to zero, so the probability tends to zero for every $\varepsilon > 0$.

Exercise 086 – Convergence in distribution to a constant

Exercise: Difficulty: ⋆ ⋆ ⋆

If $X_n \xrightarrow{d} c$, where c is constant, prove that $X_n \xrightarrow{\mathbb{P}} c$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 086 – Convergence in distribution to a constant

Solution: Detailed solution

Let $\varepsilon > 0$. Since $X_n \xrightarrow{d} c$, distribution functions converge at the continuity points of the degenerate distribution. Thus

$$\mathbb{P}(X_n \leq c - \varepsilon) \to 0,$$

and

$$\mathbb{P}(X_n \leq c + \varepsilon) \to 1.$$

Therefore

$$\mathbb{P}(|X_n - c| > \varepsilon) \leq \mathbb{P}(X_n \leq c - \varepsilon) + \mathbb{P}(X_n > c + \varepsilon) \to 0.$$

Exercise 087 – Weak law of large numbers

Exercise: Difficulty: $\star\star$

Let $X_1, \ldots, X_n$ be i.i.d. with mean μ and variance $\sigma^2 < \infty$. Use Chebyshev to show $\overline{X}_n \xrightarrow{\mathbb{P}} \mu$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 087 – Weak law of large numbers

Solution: Detailed solution

We have

$$\mathbb{E}[\overline{X}_n] = \mu, \qquad \mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n}.$$

By Chebyshev,

$$\mathbb{P}(|\overline{X}_n - \mu| > \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \to 0.$$

Thus $\overline{X}_n \to \mu$ in probability.

Exercise 088 – CLT standardization

Exercise: Difficulty: ⋆⋆⋆

Let X_i be i.i.d. with mean 10 and variance 25. Approximate $\mathbb{P}(\overline{X}_{100} > 11)$ using the central limit theorem.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 088 – CLT standardization

Solution: Detailed solution

The sample mean has mean 10 and standard deviation $5/\sqrt{100} = 0.5$. Standardize:

$$\mathbb{P}(\overline{X}_{100} > 11) = \mathbb{P}\left(\frac{\overline{X}_{100} - 10}{0.5} > 2\right).$$

By the CLT, the standardized variable is approximately standard normal. Therefore

$$\mathbb{P}(\overline{X}_{100} > 11) \approx 1 - \Phi(2).$$

Exercise 089 – Normal approximation to binomial

Exercise: Difficulty: ⋆⋆⋆

Let $X \sim \text{Bin}(400, 0.25)$. Use a normal approximation with continuity correction for $\mathbb{P}(X \leq 90)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 089 – Normal approximation to binomial

Solution: Detailed solution

The mean is $np = 100$ and the variance is $np(1-p) = 400(0.25)(0.75) = 75$. The standard deviation is $\sqrt{75}$. With continuity correction,

$$\mathbb{P}(X \leq 90) \approx \mathbb{P}\left(Z \leq \frac{90.5 - 100}{\sqrt{75}}\right).$$

Thus

$$\mathbb{P}(X \leq 90) \approx \Phi\left(\frac{-9.5}{\sqrt{75}}\right).$$

Exercise 090 – Slutsky theorem

Exercise: Difficulty: ⋆⋆⋆⋆

Suppose $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{\mathbb{P}} c$. Show what Slutsky's theorem gives for $X_n + Y_n$ and $X_n Y_n$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 090 – Slutsky theorem

Solution: Detailed solution

Slutsky's theorem says that random terms converging in probability to constants can be combined with variables converging in distribution. Therefore

$$X_n + Y_n \xrightarrow{d} X + c.$$

Also

$$X_n Y_n \xrightarrow{d} cX.$$

If $c \neq 0$, it also gives $X_n / Y_n \xrightarrow{d} X/c$ provided Y_n does not approach zero.

Exercise 091 – Continuous mapping theorem

Exercise: Difficulty: ⋆⋆⋆⋆

Let $X_n \xrightarrow{d} X$ and define $g(x) = x^2$. State and apply the continuous mapping theorem.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 091 – Continuous mapping theorem

Solution: Detailed solution

The continuous mapping theorem says that if g is continuous and $X_n \xrightarrow{d} X$, then $g(X_n) \xrightarrow{d} g(X)$. Since $g(x) = x^2$ is continuous on $\mathbb{R}$,

$$X_n^2 \xrightarrow{d} X^2.$$

The theorem would require more care if g had discontinuities at points hit by X with positive probability.

Exercise 092 – Convergence of expectations under dominated convergence

Exercise: Difficulty: ⋆⋆⋆

Let $|X_n| \leq Y$ for all n, with $\mathbb{E}[Y] < \infty$, and suppose $X_n \to X$ almost surely. What can you conclude about $\mathbb{E}[X_n]$?

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 092 – Convergence of expectations under dominated convergence

Solution: Detailed solution

The dominated convergence theorem applies because $X_n \to X$ almost surely and all X_n are dominated by one integrable random variable Y. It gives

$$\mathbb{E}[|X_n - X|] \to 0.$$

In particular, by the triangle inequality,

$$|\mathbb{E}[X_n] - \mathbb{E}[X]| \leq \mathbb{E}[|X_n - X|] \to 0.$$

Hence $\mathbb{E}[X_n] \to \mathbb{E}[X]$.

Exercise 093 – A convergence mode hierarchy

Exercise: Difficulty: ⋆⋆⋆⋆⋆

Arrange the following implications and state which converses are generally false: almost sure convergence, L^1 convergence, convergence in probability, convergence in distribution.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 093 – A convergence mode hierarchy

Solution: Detailed solution

The standard implications are

$$X_n \to X \text{ a.s.} \quad \Rightarrow \quad X_n \to X \text{ in probability} \quad \Rightarrow \quad X_n \xrightarrow{d} X.$$

Also

$$X_n \to X \text{ in } L^1 \quad \Rightarrow \quad X_n \to X \text{ in probability.}$$

In general, the converses are false. Convergence in distribution need not imply probability convergence, probability convergence need not imply almost sure convergence, and probability convergence need not imply L^1 convergence without additional assumptions.

Module 3 exam training

Characteristic functions

Exercise 094 – Delta method

Exercise: Difficulty: ⋆⋆⋆⋆⋆

Suppose $\sqrt{n}(X_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$ and g is differentiable at θ with $g'(\theta) \neq 0$. State the limiting distribution of $\sqrt{n}(g(X_n) - g(\theta))$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 094 – Delta method

Solution: Detailed solution

Use the first-order Taylor expansion

$$g(X_n) = g(\theta) + g'(\theta)(X_n - \theta) + o_p(X_n - \theta).$$

Multiplying by $\sqrt{n}$ gives

$$\sqrt{n}(g(X_n) - g(\theta)) = g'(\theta)\sqrt{n}(X_n - \theta) + o_p(1).$$

By Slutsky's theorem,

$$\sqrt{n}(g(X_n) - g(\theta)) \xrightarrow{d} \mathcal{N}(0, [g'(\theta)]^2\sigma^2).$$

Exercise 095 – Definition and simple computation

Exercise: Difficulty: ⋆⋆⋆

Let X take values -1 and 1 with probability $1/2$ each. Compute its characteristic function.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 095 – Definition and simple computation

Solution: Detailed solution

By definition,

$$\varphi_X(t) = \mathbb{E}[e^{itX}].$$

Therefore

$$\varphi_X(t) = \frac{1}{2}e^{-it} + \frac{1}{2}e^{it} = \frac{e^{it} + e^{-it}}{2} = \cos t.$$

Exercise 096 – Bernoulli characteristic function

Exercise: Difficulty: ⋆⋆⋆

Let $X \sim \text{Bern}(p)$. Compute $\varphi_X(t)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 096 – Bernoulli characteristic function

Solution: Detailed solution

Since $X = 0$ with probability $1 - p$ and $X = 1$ with probability p,

$$\varphi_X(t) = \mathbb{E}[e^{itX}] = (1-p)e^{it0} + pe^{it} = 1 - p + pe^{it}.$$

Exercise 097 – Characteristic function of a sum

Exercise: Difficulty: ⋆⋆⋆

Let X, Y be independent with characteristic functions φ_X and φ_Y. Prove that $\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 097 – Characteristic function of a sum

Solution: Detailed solution

Start from the definition:

$$\varphi_{X+Y}(t) = \mathbb{E}[e^{it(X+Y)}] = \mathbb{E}[e^{itX}e^{itY}].$$

Because X and Y are independent, e^{itX} and e^{itY} are independent. Hence

$$\mathbb{E}[e^{itX}e^{itY}] = \mathbb{E}[e^{itX}]\mathbb{E}[e^{itY}] = \varphi_X(t)\varphi_Y(t).$$

Exercise 098 – Normal characteristic function

Exercise: Difficulty: $\star\star\star$

If $Z \sim \mathcal{N}(0,1)$ has characteristic function $e^{-t^2/2}$, find the characteristic function of $X = \mu + \sigma Z$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 098 – Normal characteristic function

Solution: Detailed solution

Compute directly:

$$\varphi_X(t) = \mathbb{E}[e^{it(\mu+\sigma Z)}] = e^{it\mu}\mathbb{E}[e^{i(\sigma t)Z}].$$

Using the standard normal characteristic function at σt,

$$\mathbb{E}[e^{i(\sigma t)Z}] = e^{-\sigma^2 t^2/2}.$$

Thus

$$\varphi_X(t) = e^{it\mu-\sigma^2 t^2/2}.$$

Exercise 099 – Poisson characteristic function

Exercise: Difficulty: ⋆⋆⋆⋆

Let $X \sim \text{Poisson}(\lambda)$. Derive $\varphi_X(t)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 099 – Poisson characteristic function

Solution: Detailed solution

Use the probability mass function:

$$\varphi_X(t) = \sum_{k=0}^{\infty} e^{itk} e^{-\lambda} \frac{\lambda^k}{k!}.$$

Factor out $e^{-\lambda}$:

$$\varphi_X(t) = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^{it})^k}{k!}.$$

The series is an exponential, so

$$\varphi_X(t) = e^{-\lambda} e^{\lambda e^{it}} = \exp\{\lambda(e^{it} - 1)\}.$$

Exercise 100 – Recovering moments

Exercise: Difficulty: ⋆⋆⋆⋆

Suppose a random variable has characteristic function $\varphi(t) = e^{2it-3t^2}$. Find its mean and variance.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 100 – Recovering moments

Solution: Detailed solution

Compare with the normal characteristic function

$$e^{i\mu t - \sigma^2 t^2/2}.$$

Here $i\mu t = 2it$, so $\mu = 2$. Also

$$\frac{\sigma^2}{2} = 3,$$

so $\sigma^2 = 6$. Therefore the mean is 2 and the variance is 6.

Module 3 exam training

Conditional expectation

Exercise 101 – Using Lévy continuity theorem

Exercise: Difficulty: ⋆⋆⋆⋆⋆

Suppose $\varphi_n(t) = \left(1 + \frac{it}{n}\right)^{-n}$. Find the limiting distribution if the pointwise limit is a characteristic function.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 101 – Using Lévy continuity theorem

Solution: Detailed solution

Compute the limit:

$$\left(1+\frac{it}{n}\right)^{-n} \to e^{-it}.$$

The function e^{-it} is the characteristic function of the constant random variable -1, because $\mathbb{E}[e^{it(-1)}] = e^{-it}$. By Lévy's continuity theorem,

$$X_n \xrightarrow{d} -1.$$

Exercise 102 – Conditional expectation on a finite partition

Exercise: Difficulty: ⋆⋆⋆

Let $\Omega = \{1, 2, 3, 4\}$ with equal probabilities and $X(1) = 1$, $X(2) = 3$, $X(3) = 5$, $X(4) = 7$. Let $\mathcal{G}$ be generated by the partition $\{\{1, 2\}, \{3, 4\}\}$. Compute $\mathbb{E}[X \mid \mathcal{G}]$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 102 – Conditional expectation on a finite partition

Solution: Detailed solution

Conditional expectation is constant on each atom of $\mathcal{G}$. On $\{1,2\}$ it equals the average of 1 and 3:

$$\frac{1+3}{2} = 2.$$

On $\{3,4\}$ it equals the average of 5 and 7:

$$\frac{5+7}{2} = 6.$$

Thus $\mathbb{E}[X \mid \mathcal{G}] = 2$ on $\{1,2\}$ and 6 on $\{3,4\}$.

Exercise 103 – Tower property

Exercise: Difficulty: ⋆⋆⋆

Let $\mathcal{H} \subseteq \mathcal{G} \subseteq \mathcal{F}$. State the tower property and explain its meaning.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 103 – Tower property

Solution: Detailed solution

The tower property says

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}].$$

Meaning: if we first average X using the information $\mathcal{G}$ and then average again using the coarser information $\mathcal{H}$, the final result is the same as averaging directly with respect to $\mathcal{H}$.

Exercise 104 – Taking out what is known

Exercise: Difficulty: ⋆⋆⋆⋆

Let Y be $\mathcal{G}$-measurable and bounded. Prove the identity $\mathbb{E}[YX \mid \mathcal{G}] = Y\mathbb{E}[X \mid \mathcal{G}]$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 104 – Taking out what is known

Solution: Detailed solution

The idea is that Y is already known when conditioning on $\mathcal{G}$. To verify the defining property, take any $A \in \mathcal{G}$. Then

$$\int_A Y\,\mathbb{E}[X \mid \mathcal{G}]\,d\mathbb{P}$$

equals

$$\int_A YX\,d\mathbb{P},$$

first for simple Y by splitting into atoms or level sets, and then for bounded measurable Y by approximation. Hence $Y\mathbb{E}[X \mid \mathcal{G}]$ satisfies the defining property of $\mathbb{E}[YX \mid \mathcal{G}]$.

Exercise 105 – Conditioning on a random variable

Exercise: Difficulty: $\star\star\star$

Let X, Y be discrete. Explain how to compute $\mathbb{E}[X \mid Y]$ from the conditional probabilities of X given $Y = y$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 105 – Conditioning on a random variable

Solution: Detailed solution

For each possible value y with $\mathbb{P}(Y = y) > 0$, compute

$$g(y) = \mathbb{E}[X \mid Y = y] = \sum_x x\mathbb{P}(X = x \mid Y = y).$$

Then

$$\mathbb{E}[X \mid Y] = g(Y).$$

Thus conditional expectation given Y is a random variable obtained by applying a deterministic function g to Y.

Exercise 106 – Projection interpretation

Exercise: Difficulty: ⋆⋆⋆⋆

In L^2, explain why $\mathbb{E}[X \mid \mathcal{G}]$ can be interpreted as the best approximation of X using only $\mathcal{G}$-measurable variables.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 106 – Projection interpretation

Solution: Detailed solution

The space of square-integrable $\mathcal{G}$-measurable random variables is a closed subspace of L^2. Conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$ is the orthogonal projection of X onto this subspace. Therefore it minimizes

$$\mathbb{E}[(X - Z)^2]$$

over all square-integrable $\mathcal{G}$-measurable Z. The error $X - \mathbb{E}[X \mid \mathcal{G}]$ is orthogonal to every such Z.

Exercise 107 – Conditional expectation and independence

Exercise: Difficulty: ⋆⋆⋆

If X is integrable and independent of $\mathcal{G}$, prove that $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X]$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 107 – Conditional expectation and independence

Solution: Detailed solution

The candidate is the constant random variable $\mathbb{E}[X]$. For any $A \in \mathcal{G}$, independence gives

$$\mathbb{E}[X\mathbf{1}_A] = \mathbb{E}[X]\mathbb{P}(A).$$

But

$$\int_A \mathbb{E}[X]d\mathbb{P} = \mathbb{E}[X]\mathbb{P}(A).$$

Therefore the constant $\mathbb{E}[X]$ satisfies the defining property of conditional expectation.

Exercise 108 – Conditional variance decomposition

Exercise: Difficulty: ⋆⋆⋆⋆

State and prove the identity $\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid \mathcal{G})] + \mathrm{Var}(\mathbb{E}[X \mid \mathcal{G}])$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 108 – Conditional variance decomposition

Solution: Detailed solution

Use $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$. Conditional variance is

$$\mathrm{Var}(X \mid \mathcal{G}) = \mathbb{E}[X^2 \mid \mathcal{G}] - (\mathbb{E}[X \mid \mathcal{G}])^2.$$

Taking expectations gives

$$\mathbb{E}[\mathrm{Var}(X \mid \mathcal{G})] = \mathbb{E}[X^2] - \mathbb{E}[(\mathbb{E}[X \mid \mathcal{G}])^2].$$

Also

$$\mathrm{Var}(\mathbb{E}[X \mid \mathcal{G}]) = \mathbb{E}[(\mathbb{E}[X \mid \mathcal{G}])^2] - (\mathbb{E}[X])^2.$$

Adding the two formulas gives $\mathrm{Var}(X)$.

Exercise 109 – Least squares predictor

Exercise: Difficulty: ⋆⋆⋆⋆⋆

Let X, Y be square-integrable and suppose $\mathbb{E}[X \mid Y] = a + bY$. Show that this is the best mean-square predictor among all functions of Y.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 109 – Least squares predictor

Solution: Detailed solution

By the projection property of conditional expectation, for every square-integrable function $h(Y)$,

$$\mathbb{E}[(X - \mathbb{E}[X \mid Y])^2] \leq \mathbb{E}[(X - h(Y))^2].$$

If $\mathbb{E}[X \mid Y] = a + bY$, then the best predictor among all measurable functions of Y is exactly the linear expression $a + bY$. This is stronger than being best only among linear functions.

Exercise 110 – Computing a conditional expectation from a density

Exercise: Difficulty: ⋆⋆⋆⋆⋆

Let $f(x,y) = 2$ on $0 < x < y < 1$ and 0 otherwise. Compute $\mathbb{E}[X \mid Y = y]$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 110 – Computing a conditional expectation from a density

Solution: Detailed solution

First find the marginal density of Y:

$$f_Y(y) = \int_0^y 2\,dx = 2y, \qquad 0 < y < 1.$$

Thus the conditional density of X given $Y = y$ is

$$f_{X\mid Y=y}(x) = \frac{2}{2y} = \frac{1}{y}, \qquad 0 < x < y.$$

So conditionally on $Y = y$, X is uniform on $(0, y)$. Therefore

$$\mathbb{E}[X \mid Y = y] = \frac{y}{2}.$$

Exercise 111 – Exam synthesis: conditional expectation and variance

Exercise: Difficulty: ⋆⋆⋆⋆⋆

Let $Y \sim \text{Bern}(0.4)$ and, conditionally on Y, let $X \mid Y = 0$ have mean 1 and variance 2, while $X \mid Y = 1$ has mean 4 and variance 3. Compute $\mathbb{E}[X]$ and $\text{Var}(X)$.

Method: What to identify before solving

Object type, formula to use, and assumptions required.

Solution 111 – Exam synthesis: conditional expectation and variance

Solution: Detailed solution

First use total expectation:

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]] = 1 \cdot 0.6 + 4 \cdot 0.4 = 0.6 + 1.6 = 2.2.$$

Next use total variance:

$$\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(\mathbb{E}[X \mid Y]).$$

The first term is

$$2(0.6) + 3(0.4) = 2.4.$$

The conditional mean equals 1 with probability 0.6 and 4 with probability 0.4, so

$$\mathrm{Var}(\mathbb{E}[X \mid Y]) = 0.6(1 - 2.2)^2 + 0.4(4 - 2.2)^2 = 0.864 + 1.296 = 2.16.$$

Thus

$$\mathrm{Var}(X) = 2.4 + 2.16 = 4.56.$$

Final checklist for Module 3 exam exercises

Method: Before submitting a solution

Check the following points.

1. Did you define the event or random variable clearly?
2. Did you use conditioning only when the conditioning event has positive probability?
3. Did you distinguish independence from zero covariance?
4. Did you check the support of each random variable?
5. Did you state the convergence mode precisely?
6. Did you justify each theorem used: Jensen, Markov, Chebyshev, LLN, CLT, Slutsky, dominated convergence, tower property?